Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/11/2022
PI Name:	Bell, Suzanne Ph.D.		
Project Title:	Generalizability of Operational Feasibility and Acceptability of HFBP Exploration Measures in Nezemnyy Eksperimental'nyy Kompleks (NEK)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2021 at time of Task Book entry: In 2018 Peter Roma was the Principal Investigator (PI), CoIs were Lauren Landon, Sandra Whitmire. In 2020 PIs were Suzanne Bell and Peter Roma and CoIs were Lauren Landon, Gloria Leon, Erin Flynn-Evans. In 2021 PI is Suzanne Bell (NASA), with CoIs: Lauren Landon (KBR), Gloria Leon (Independent Consultant), and Erin Flynn-Evans (NASA). March 2022 addition: Sheena Dev Ph.D. was added as Co-I in May 2021.		
COI Name (Institution):	Landon, Lauren Ph.D. (KBR/NASA Johnson Space Center) Leon, Gloria Ph.D. (Independent Consultant) Flynn-Evans, Erin Ph.D. (NASA Ames Research Center) Dev, Sheena (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Given the small sample sizes associated with space and analog research, it is critical that comparable data using the same set of measures on key Behavioral Health and Performance (BHP) variables are collected across participants.
The purpose of this task is to establish, evaluate, and manage a common or “standard measures,” called Human Factors Behavioral Performance -- Exploration Measures (HFBP-EM) for use in spaceflight and analog research. These measures will be used to develop baselines, systematically characterize risk likelihood and consequences, and assess effectiveness of countermeasures for human factors and behavioral performance risk factors. This task allows these measures to be consistently implemented with evidence-based iterations over time, allows comparability with other HFBP-EM applications, and data sharing with Nezemnyy Eksperimental’nyy Kompleks (NEK) investigators as needed to inform Human Research Program (HRP)’s research gaps.

The proposed research is a supplement to the previous directed task of developing and evaluating a standardized suite of measures that can capture space-relevant behavioral health and performance phenomena across spaceflight and analog environments (called HFBP Exploration Measures or HFBP-EM, formerly Behavior Health and Performance [BHP] Standard Measures). The current proposal includes HFBP-EM in the Scientific International Research In a Unique terrestrial Station-21 (SIRIUS) mission and augments the protocol with a measure of habitability and Russian language personality, affective, and group dynamics measures. To date, most HFBP-EM data have been collected in the Human Exploration Research Analog (HERA) at Johnson Space Center. These missions are up to 45 days and include US participants. Data collection in the SIRIUS missions provides an important data collection opportunity for a longer-duration mission (i.e., 8 month), in a high fidelity mission simulation analog (i.e., NEK) with habitability features different from HERA, and from a multinational crew.

The differences in habitat (i.e., NEK versus HERA) allow for testing of the Spaceflight Habitability Acceptability Questionnaire (SHAQ). SHAQ measures the BHP impacts of net habitable volume and layout, which allows for additional risk coverage (Human Research Roadmap gap HSIA 101). SHAQ will allow for comprehensive and systematic evaluation of existing operational habitats (e.g., International Space Station, spaceflight analogs), habitats in development (e.g., Orion, Gateway), and multiple prototypes across mission types (e.g., lunar surface, Mars transit, Mars surface) to inform habitat designs, recommendations, and requirements.

Our research extends previous HFBP-EM efforts by including Russian translated measures of personality, affect, and group dynamics that have been collected previously in analog environments. This will allow us to further examine the feasibility and acceptability of the HFBP-EM suite, as compared to other measures taken in the preferred language of the participant. It will also increase the scientific yield in NEK.

Aims:

1. Provide a set of BHP standard measurements for investigators to use in proposed projects.

2. Enable comparison of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data.

3. Provide database for data-mining and integrative modeling and increase research data quality and transfer to Life Sciences Data Archive (LSDA).

Specific Aims:

The specific aims of our research are to:

Aim 1: Assess Operational Feasibility of HFBP-EM in the Russian NEK space analog.

Aim 2: Assess Operational Acceptability of HFBP-EM in the Russian NEK space analog.

Aim 3: Provide a set of HFBP-EM for data-sharing with intramural and extramural investigators to use in proposed projects.

Aim 4: Enable comparisons of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data.

Aim 5: Contribute to HFBP-EM database for data-mining and integrative modeling and increase research data quality and transfer to LSDA.

Aim 6: Provide an assessment of behavioral and health performance impacts of habitat layout and acceptability.

Aim 7: Provide a comparative assessment of individual and crew functioning in isolation with measures taken in the participants’ preferred language with those collected in English.

Aim 8: Examine the extent to which a monthly retrospective sleep quality index relates to daily subjective and objective sleep measures, as well as subjective ratings of neurobehavioral functioning and mood over time.

This task will contribute to gap closure by facilitating all tasks using the Institute of Biomedical Problems (IBMP) NEK Chamber analog in the risk areas of "Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders" (BMed), "Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload" (Sleep), and "Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team" (Team).

Research Methods:

Data will be collected from the Ground-Based Experiment Complex (NEK) space analog facility at the Russian Institute for Biomedical Problems (IBMP). Pre-mission data will be collected from all individuals (primary crew and backups).

Rationale for HRP Directed Research:	This proposal qualifies for a directed study due to the urgent, time-sensitive need to provide “standard measures” as the foundation to achieve consistent research measures for data-sharing in NEK and to meet the highly constrained, operationally-focused data gathering and analysis that allows for greater consistency in the research methods that are very specific to NASA Human Research Program (HRP) standard measures development. Timelines for integration of study hardware and protocol are further constrained due to international agreements, Export Control, and other requirements that modify/accelerate schedules. Completion of the required research, support to both analog and operational requirements, and the required vetting of the evidence-based standards makes the solicitation process prohibitive. Moreover, the projected start date of the first SIRIUS study is rapidly approaching. In 2020: The proposed research is a supplement to the previous directed task of developing and evaluating a standardized suite of measures that can capture space-relevant behavioral health and performance phenomena across spaceflight and analog environments (called HFBP Exploration Measures or HFBP-EM, formerly Behavior Health and Performance [BHP] Standard Measures). The current proposal includes HFBP-EM in the SIRIUS-21 mission and augments the protocol with a measure of habitability and Russian language personality, affective, and group dynamics measures. This project qualifies for a directed task because of: (a) the need for NASA to ensure continuity across all HFBP-EM data collection; (b) the importance of providing data to other internal and external researchers in a timely manner; and (c) the ability for the lab to report quickly on data collection success and barriers. Further, the time sensitive and urgent need to develop a validated standard set of measures that can be used across laboratory, analogs, field, and spaceflight, further qualifies this research as a directed study.
Research Impact/Earth Benefits:	
Task Progress:	This entry reports on task progress from 8/1/2017 to 3/15/2022. During this time, the Behavioral Health & Performance (BHP) Laboratory has made tremendous progress in three primary areas: (1) Human Factors Behavioral Performance - Exploration Measures (HFBP-EM) data collection in the NEK-SIRIUS17 and SIRIUS19 spaceflight analog missions; (2) processing of HFBP-EM data in SIRIUS17 to the extent needed for the pilot test; (3) processing of HFBP-EM data collected in the NEK-SIRIUS19 spaceflight analog mission; and (4) protocol development and data collection in the NEK-SIRIUS 21 spaceflight analog mission. [Ed. Note: Scientific International Research In a Unique terrestrial Station (SIRIUS)]. HFBP-EM data collection was first implemented in the NEK-SIRIUS17 mission that concluded on 12/1/2017. This was a short 17-day mission that served as a pilot test to assess the feasibility and acceptability for subsequent longer duration analog studies in NEK such as the 4-month SIRIUS19 and the ongoing 8-month SIRIUS21 missions described below. Data yield was inconsistent across measures with percent usability ranging from 0% to 100%. For example, team performance as rated by mission control was entirely missing and actigraphy data was collected from two of the six crew members. The pattern of received survey responses for some, but not all, individuals was indicative of a language barrier, non-compliance, or careless responding. ROBoT-r, an important operationally relevant performance task, was administered on hand controllers that were not equivalently designed nor calibrated to the other ROBoT systems utilized in the other analogs. While SIRIUS17 served as an important pilot to prepare for data collection in longer missions in NEK, the data are not considered usable data for HFBP-EM scientific aims such as summarizing the trajectory of behavioral health and performance in individuals and teams in an isolated and confined environment. Thus, the data were processed only to the extent needed for the pilot. The study team learned critical lessons about the implementation and feasibility of HFBP-EM during this mission which directly resulted in very successful data collection in the longer duration missions that followed. The SIRIUS17 data should not be processed further or considered further beyond these lessons learned. HFBP-EM data collection in the 4-month NEK-SIRIUS19 mission concluded on 07/31/2019 with n = 6 participants. Data yield was very high. Objective performance data yield (Cognition and ROBoT): 96%, Wearables data yield (actigraphy and heart rate): 97.5%; Survey data yield 97.6%. Data cleaning, processing, and analysis of HFBP-EM collected during the NEK-SIRIUS19 mission has been completed. Careless responses on surveys were identified and flagged. Data from surveys which were originally developed and validated with Likert response formats but had been collected on visual analog scales were transformed and combined into subscales where applicable. Actigraphy and heart rate data were processed according to best practices, and sleep and heart rate metrics were summarized as daily aggregates. Objective performance tasks were subjected to data transformations and key performance metrics were extracted. A set of meta-data variables (e.g., mission, participant ID, mission day) were generated to define key variables with which data could be merged across missions for future analyses. Finally, we worked with NASA Research Operations & Integration (ROI) to prepare for the 8-month SIRIUS21 mission. Data collection is underway since the start of the pre-mission phase on 09/21/2021. Mid-mission data yield is forthcoming and will be provided in our mid-mission brief report expected due in April 2022. In the coming year, we will continue to collect data for the remainder of the 8-month mission, as well as prepare, process, and summarize data.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
Abstracts for Journals and Proceedings	Bell ST, Dev SI, Whiting SE, Miller JCW, Khader AM, Roma P. "Human Factors and Behavioral Performance Exploration Measures harmonized across HERA, NEK, and ISS: Behavioral medicine risk. " 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Bell ST, Miller JCW, Landon LB, Dev SI, Roma P. "Human Factors and Behavioral Performance Exploration Measures harmonized across HERA, NEK, and ISS: Teams risk. " 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Bell ST, Miller JDW, Whiting SE, Landon LB, Dev SI, Begerowski SR, Doerr A, Roma P. "Development and initial validation of Human Factors and Behavioral Performance Exploration Measures (HFBP-EM) short-form surveys." 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022

Abstracts for Journals and Proceedings	Bell ST, Roma P, Whiting S, Landon L. "Summary of Human Factors Behavioral Performance Exploration Measures in SIRIUS'19. " 23rd International Academy of Astronomics (IAA) Humans in Space Symposium, Virtual, Moscow, Russia, April 5-8, 2021. Abstracts. 23rd International Academy of Astronomics (IAA) Humans in Space Symposium, Virtual, Moscow, Russia, April 5-8, 2021. , Apr-2021